

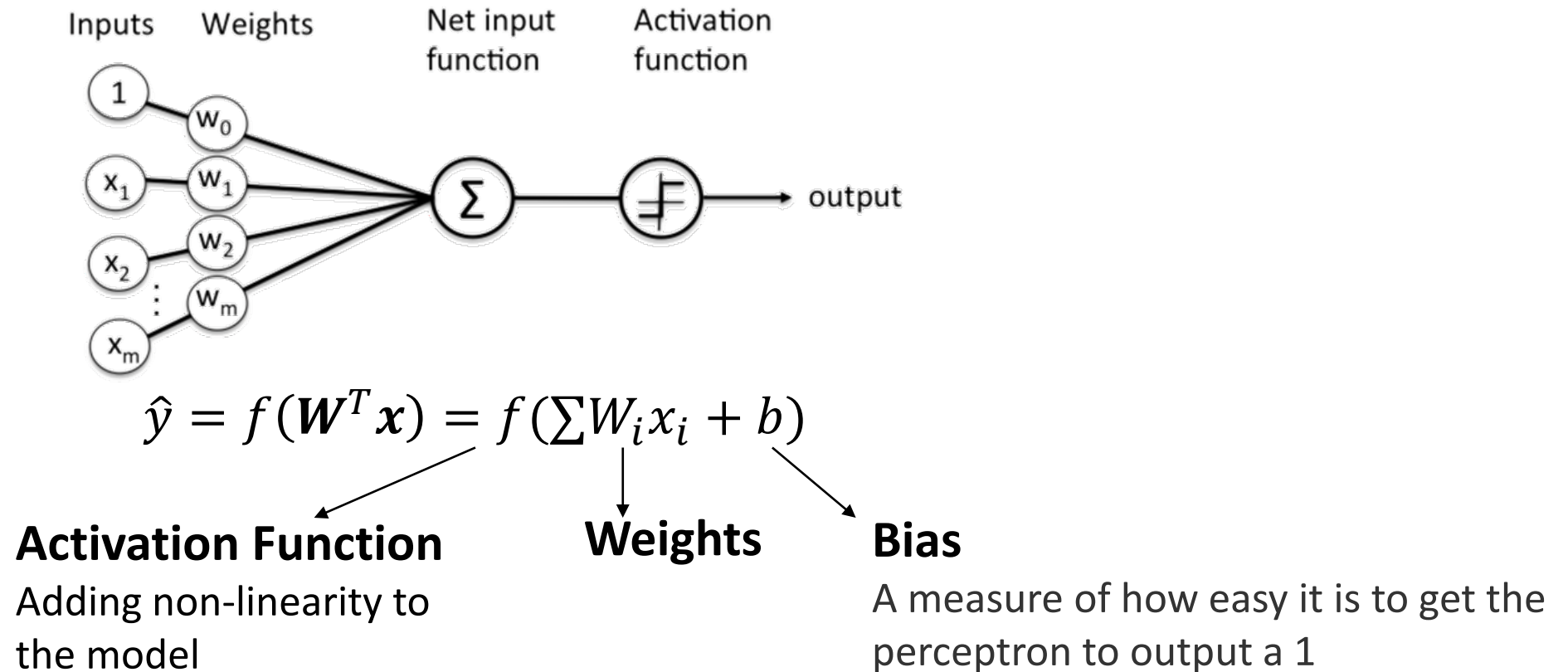
Module 7. Neural Networks

- ❑ Basic Concepts
- ❑ Perceptron Learning 
- ❑ Artificial Neural Networks
- ❑ Backpropagation
- ❑ Summary

Train a Neuron to Fire

- ❑ In the training mode, the neuron can be trained to fire (or not), for particular input patterns.
- ❑ In the using mode, when a taught input pattern is detected at the input, its associated output becomes the current output. If the input pattern does not belong in the taught list of input patterns, the firing rule is used to determine whether to fire or not.
- ❑ The firing rule is an important concept in neural networks and accounts for their high flexibility. A firing rule determines how one calculates whether a neuron should fire for any input pattern. It relates to all the input patterns, not only the ones on which the node was trained on previously.

Perceptron: Predecessor of a Neural Network



- ❑ Computes a weighted sum of inputs
- ❑ Invented in 1957 by Frank Rosenblatt. The original perceptron model does not have a non-linear activation function